

1-zero:

$$X_{24} = X_{14} - X_{13}$$

⋮

$$X_{2n-1} = X_{1n-1} - X_{13}$$

$$X_{2n} = -X_{13}$$

$$\begin{array}{c} X_{35} \quad X_{36} \quad \dots \quad X_{3n} \\ X_{46} \quad \dots \quad X_{4n} \\ \vdots \\ X_{n-2,n} \end{array}$$

2-zero

$$X_{35} = X_{15} - X_{14}$$

⋮

$$X_{3n-1} = X_{1n-1} - X_{14}$$

$$X_{3n} = -X_{14}$$

$$X_{13}$$

$$X_{46} \quad \dots \quad X_{4n}$$

$$\vdots$$

$$X_{n-2,n}$$

also: $X_{2j} = X_{3j} + X_{2n} \quad j = 5, \dots, n$

closed enclosed in $(k+1)$ -gon

3-zero

$$X_{46} = X_{16} - X_{15}$$

⋮

$$X_{4n-1} = X_{1n-1} - X_{15}$$

$$X_{4n} = -X_{15}$$

$$X_{2j} = X_{4j} + X_{25} \quad j = 6, \dots, n$$

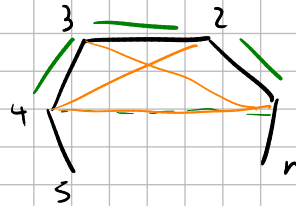
$$X_{3j} = X_{4j} + X_{35} \quad j = 6, \dots, n$$

$$\begin{array}{c} X_{13} \quad X_{14} \\ X_{24} \end{array}$$

$$X_{57} \quad \dots \quad X_{5n}$$

$$\vdots$$

$$X_{n-2,n}$$



k-zero

$$X_{k+1, k+3} = X_{1, k+3} - X_{1, k+2}$$

⋮

$$X_{k+1, n-1} = X_{1, n-1} - X_{1, k+2}$$

$$X_{k+1, n} = -X_{1, k+2}$$

$$X_{ij} = X_{k+1, j} + X_{2, k+2} \quad \begin{array}{l} i = 2, \dots, k \\ j = k+3, \dots, n \end{array}$$

free:

enclosed in $(k+1)$ -gon

$$\begin{array}{c} X_{13} \quad X_{14} \quad \dots \quad X_{1, k+1} \\ X_{24} \quad \dots \quad X_{2, k+1} \\ \vdots \\ X_{k-1, k+1} \end{array}$$

$$X_{k+2, k+4} \quad X_{k+2, k+5} \quad \dots \quad X_{k+2, n}$$

$$X_{k+3, k+5} \quad \dots \quad X_{k+3, n}$$

$$\vdots$$

$$X_{n-2, n}$$

we choose sets of free things for each k

by taking union of "born free" variables and choosing one

each from each pair $(X_{k+1, j}, X_{1, j}) \quad j = k+3, \dots, n-1$

what survives the k -zero?

for each set of k adjacent edges, if we look at the $(k+1)$ fan bordered by closing it, then at least one of 2 things happens:

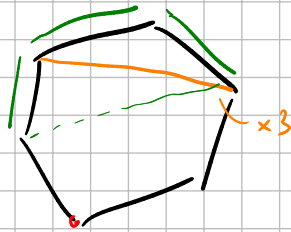
(P1) • there is an enclosed vertex that crosses out of the $(k+1)$ -fan
(alternatively there is a chord in the diagram that crosses the $(k+1)$ -fan)



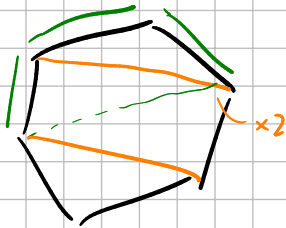
(P2) • the $(k+1)$ -fan is contained in a $(k+2)$ -fan formed by edges + chords.

anything that doesn't satisfy one of the 2 points is killed

ex:

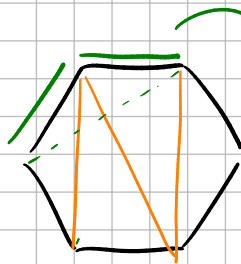


does not survive $k=3$



is part of 5-fan

triangulation ex:

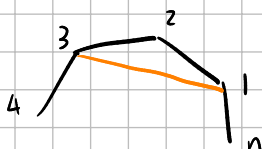


these edges are non part of a 4-fan
but there chords that cross the $(k+1)$ fan
so it's ok

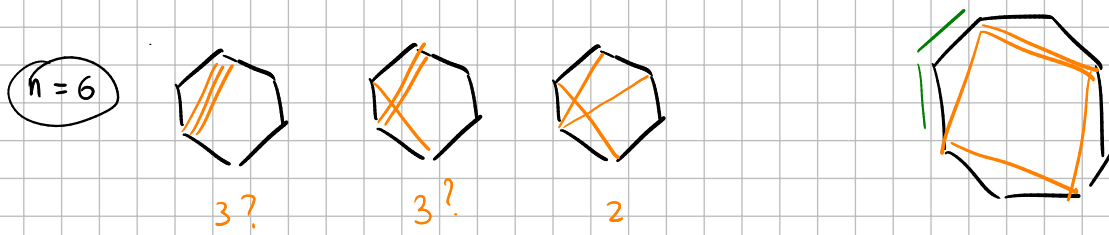
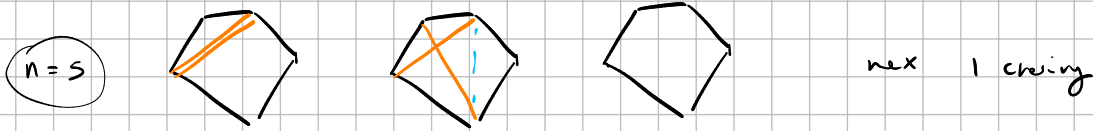
Point (1) is because we send the $X_{2i}, X_{3i}, \dots, X_{ki}$ $i = k+2, \dots, n$
to ∞ so the terms that have the things that cross out go to zero.

look at crossings

start crossing itself (appears 2+ times)



What's max # of crossings we can have with $n-3$ chords?



one approach: consider triangulations as subsets of sets of chords

start with $S = \{X_{13}, X_{14}, \dots\}$ all chords.

look at all multisets of order $n-3$. These are all the diagrams we consider.

Reformulate our problem as a characterization of triangulations:

• diagram is triangulation iff there are no crossings

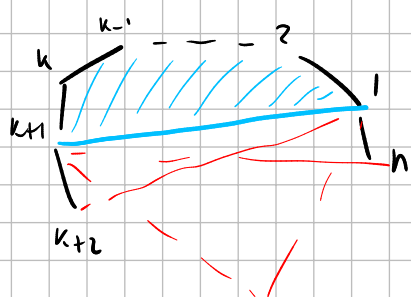
② triangulations are the only diagrams s.t. for each consecutive k edges, either prop ① or ② holds

$\checkmark p_1, p_2$ hold for all k

① $\Leftrightarrow$ ② $\leftarrow$ diagram is a triangulation

Prop. any triangulation satisfies the property that for each k adjacent edges, either the $(k+1)$ -gon has outside crossings or it is part of $(k+2)$ -gon.

proof look at edges $(12), (23), \dots, (k, k+1)$. Suppose that the $(k+1)$ -gon is self-contained (no external crossings). Then adding the chord $(1, k+1)$ to the triangulation does not introduce any crossing. This new $(1, 1+k)$ must be part of the triangulation, otherwise we would have a triangulation with $(n-2)$ chords. Since this is a triangulation, $(1, k+1)$ must have a triangle outside of the $(k+1)$ -gon.

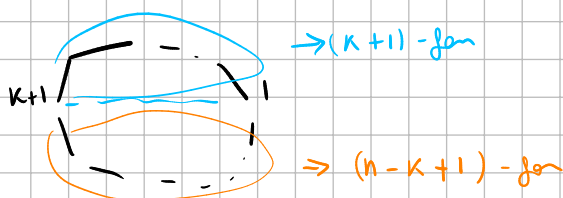


- so we showed that either
- the $(k+1)$ -gon has outside crossings
 - the $(k+1)$ -gon is part of a $(k+2)$ -gon

for a triangulation: if there is a chord that we can cut along to, then it belongs to the triangulation.



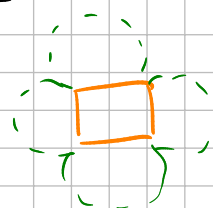
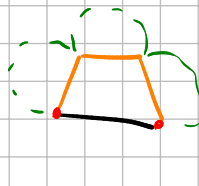
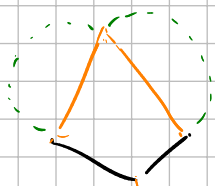
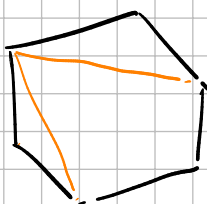
try k under assumption that $(k+1)$ -gon is self contained



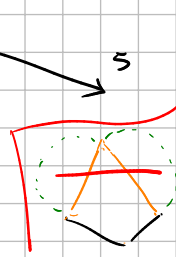
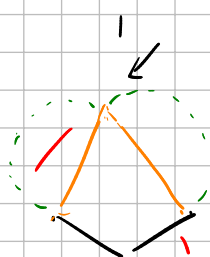
suppose that they are both self-contained. prove that something breaks if not a triangulation

start by showing that if $(k+1)$ -gon does not have enough inner chords to be a triangulation then diagram does not survive.

what happens if a $(k+1)$ -gon has less than $k-2$ chords?



add crossing chord!

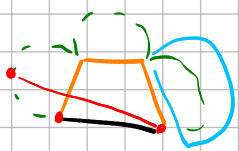
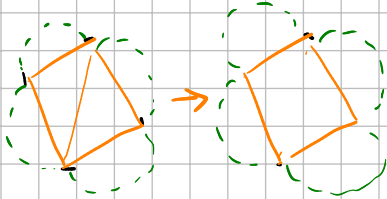


does not survive as the bottom edges are not part of triangle.
($k=1$ null)

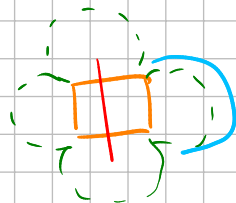
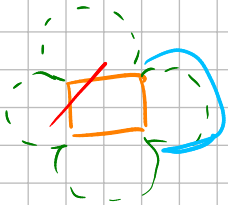
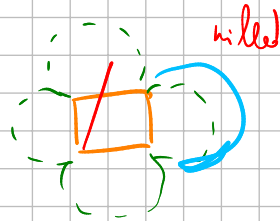
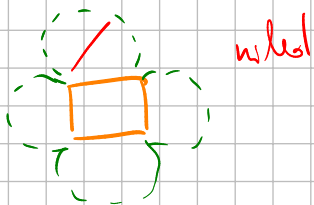
$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 ok don't survive $k=1$



- faces that are 1 step removed from Δ s are killed by some k -zero.

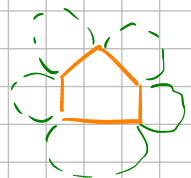
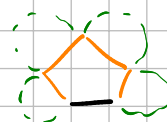


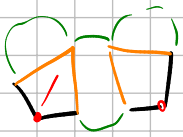
look at these edges: if they are k edges in blue region
 this is killed by k -zero because it is not in $(k+2)$ -gen



remove 2 things

polygons:





not here one chessing per region

↓ reduces

